

Final Project

Airbnb Price Prediction in Tokyo

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Machine Learning, SoSe 2026

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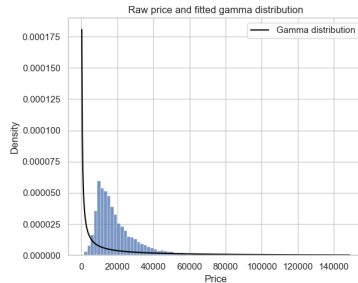
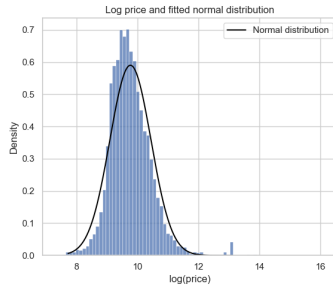
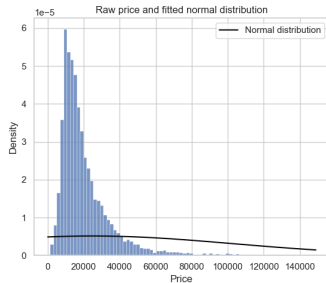
1. Data Cleaning and Preparation
2. Feature Engineering
3. Modeling and Tuning
4. Analysis and Reporting

1. Data Cleaning and Preparation

1.1 Data

- ▶ City: Tokyo
- ▶ Inside Airbnb snapshot: September 2025
- ▶ Raw data: 27,945 listings
- ▶ Files: listings, reviews, images
- ▶ Modalities: tabular, spatial, text, images
- ▶ Target: nightly asking price

1.2 Price Distribution



1.3 Data Preparation

- ▶ Price cleaning
- ▶ Number of positive prices: 25,480
- ▶ Mean price: 24,782 JPY
- ▶ Median price: 16,200 JPY
- ▶ Log transformation
- ▶ Train-test split: 80/20

2. Feature Engineering

2.1 Tabular Features

Numeric features

- ▶ Median imputation
- ▶ Log transforms
- ▶ Standardization

Categorical features

- ▶ Missing categories
- ▶ One-hot encoding
- ▶ Review-score buckets
- ▶ Amenity indicators: 21

2.2 Spatial Features and Interactions

- ▶ Latitude and longitude splines
- ▶ Neighbourhood $\times$ room type
- ▶ Property group $\times$ accommodates
- ▶ Amenities $\times$ property group
- ▶ Review score $\times$ review count
- ▶ Final matrix: 1,017 features

2.3 Text Features

- ▶ Listing text: title, description, neighbourhood overview, host profile
- ▶ Review text: 10 latest reviews
- ▶ Amenities: text representation
- ▶ Native CatBoost text
- ▶ Word and character TF-IDF
- ▶ Truncated SVD: 128 features

2.4 Image Features

Transformation

- ▶ Main listing image
- ▶ CLIP image embedding
- ▶ Similarity to 15 prompts

Model features

- ▶ 15 similarity scores
- ▶ Image availability indicator

3. Modeling and Tuning

3.1 Linear Regression

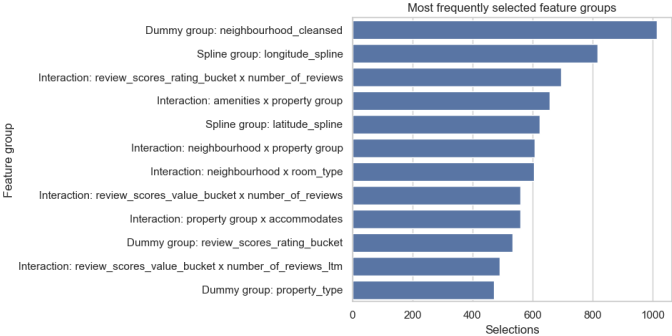
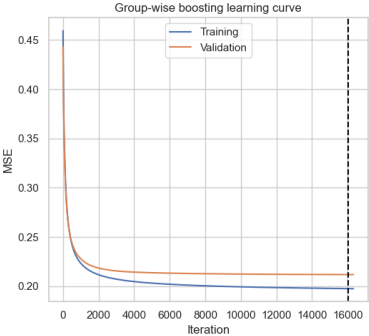
Input

- ▶ Tabular features
- ▶ Spatial features

Results

- ▶ Log-price MSE: 0.202
- ▶ Log-price R^2 : 0.550
- ▶ Price MAE: 8,960 JPY
- ▶ Median AE: 3,387 JPY

3.2 Group-wise Linear Boosting



3.2 Group-wise Linear Boosting

Input

- ▶ Tabular features
- ▶ Spatial features

Results

- ▶ Log-price MSE: 0.198
- ▶ Log-price R^2 : 0.558
- ▶ Price MAE: 8,941 JPY
- ▶ Median AE: 3,458 JPY

3.3 Histogram Gradient Boosting

Input

- ▶ Tabular features
- ▶ Spatial features

Results

- ▶ Log-price MSE: 0.142
- ▶ Log-price R^2 : 0.683
- ▶ Price MAE: 8,010 JPY
- ▶ Median AE: 2,765 JPY

3.4 CatBoost: Tabular Data

Input

- ▶ Tabular features
- ▶ Spatial features

Results

- ▶ Log-price MSE: 0.117
- ▶ Log-price R^2 : 0.739
- ▶ Price MAE: 6,904 JPY
- ▶ Median AE: 2,382 JPY

3.5 CatBoost: TF-IDF reduced with SVD

Input

- ▶ Tabular features
- ▶ Spatial features
- ▶ TF-IDF reduced with SVD

Results

- ▶ Log-price MSE: 0.113
- ▶ Log-price R^2 : 0.748
- ▶ Price MAE: 6,293 JPY
- ▶ Median AE: 2,377 JPY

3.6 CatBoost: Native Text

Input

- ▶ Tabular features
- ▶ Spatial features
- ▶ Native text features

Results

- ▶ Log-price MSE: 0.105
- ▶ Log-price R^2 : 0.766
- ▶ Price MAE: 6,230 JPY
- ▶ Median AE: 2,159 JPY

3.7 CatBoost: Native Text and CLIP

Input

- ▶ Tabular features
- ▶ Spatial features
- ▶ Native text features
- ▶ CLIP prompt features

Results

- ▶ Log-price MSE: 0.105
- ▶ Log-price R^2 : 0.765
- ▶ Price MAE: 6,239 JPY
- ▶ Median AE: 2,202 JPY

3.8 CatBoost: Native Text and CLIP, Long Run

Input

- ▶ Tabular features
- ▶ Spatial features
- ▶ Native text features
- ▶ CLIP prompt features

Results

- ▶ Log-price MSE: 0.102
- ▶ Log-price R^2 : 0.772
- ▶ Price MAE: 6,053 JPY
- ▶ Median AE: 2,133 JPY

3.9 CatBoost: Native Text, Long Run

Input

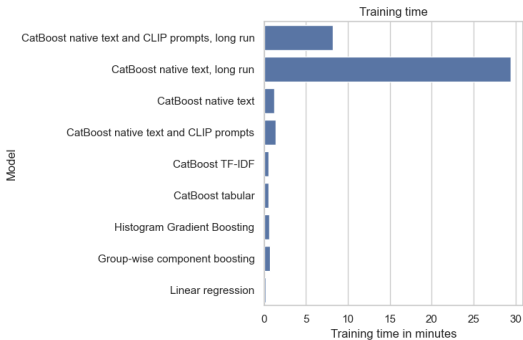
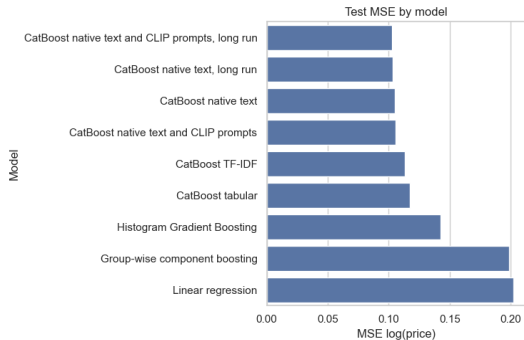
- ▶ Tabular features
- ▶ Spatial features
- ▶ Native text features

Results

- ▶ Log-price MSE: 0.103
- ▶ Log-price R^2 : 0.770
- ▶ Price MAE: 5,929 JPY
- ▶ Median AE: 2,081 JPY

4. Analysis and Reporting

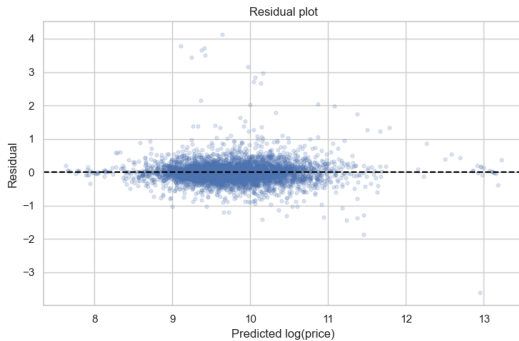
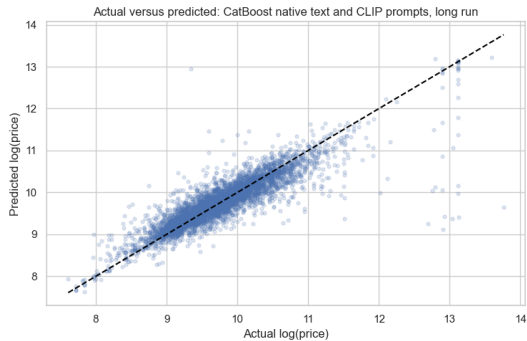
4.1 Model Comparison



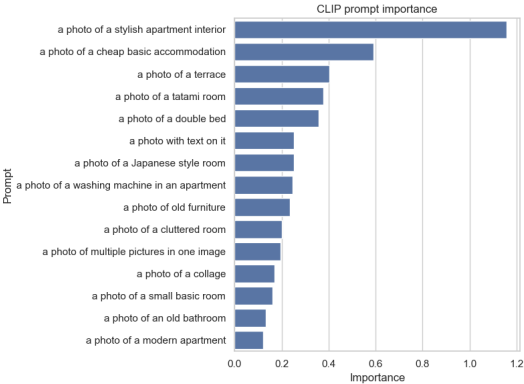
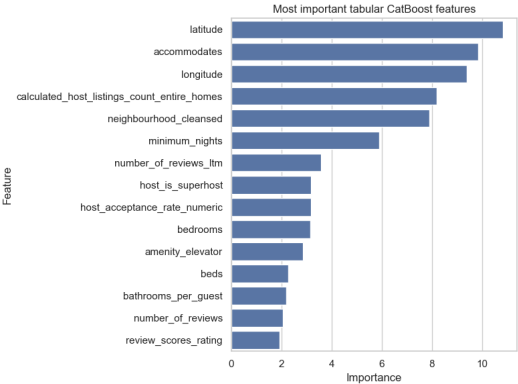
4.1 Main Results

- ▶ Best log-price MSE: Native text and CLIP, long run
- ▶ Lowest price MAE: Native text, long run
- ▶ Main improvement: text features
- ▶ No clear improvement: image features

4.2 Predictions and Residuals



4.3 Feature Importance



4.3 Main Features

- ▶ Location
- ▶ Capacity
- ▶ Host activity
- ▶ Minimum nights

4.4 Limitations

- ▶ Asking prices
- ▶ Single point-in-time snapshot
- ▶ Extreme round prices
- ▶ Missing reviews: new listings
- ▶ Repeated images and descriptions: same host